SHIV NADAR UNIVERSITY CHENNAI

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SCHOOL OF ENGINEERING
CS4801 DevOps Lab
CONTINUOUS EVALUATION EXAMINATION QUESTION PAPER

CI/CD PIPELINE LAB ASSESSMENT

Git • Maven • JUnit 5 • Jenkins Declarative Pipeline

Scenario

You are a DevOps engineer maintaining a Java/Maven Course Registration service. Your task is to improve the code and tests, manage the work through Git, and define a Jenkins Declarative Pipeline that automates the build and test workflow.

General instructions

- Start from the given Student-Starter baseline repository. Do not replace it with a different project.
- Work in your own local Git repository. Use a feature branch and meaningful commits; preserve a readable history.
- Inspect the code and existing tests before making changes. Do not delete or weaken existing tests merely to make the build pass.
- Use Maven to compile, test, and package the application. Investigate failures rather than reporting only the final status.
- Use JUnit 5 for new or modified tests. Tests must assert behaviour and state, not merely that a method runs.
- Write a Jenkins Declarative Pipeline in a file named Jenkinsfile. A live Jenkins execution is not required unless access to a shared Jenkins instance is explicitly provided.
- Use `bat` steps for a Windows agent or `sh` steps for a Linux agent, and state which environment your Jenkinsfile targets.
- Do not submit fabricated screenshots, console logs, or claims of a Jenkins run that did not occur. If Jenkins is unavailable, explain how the pipeline would run.

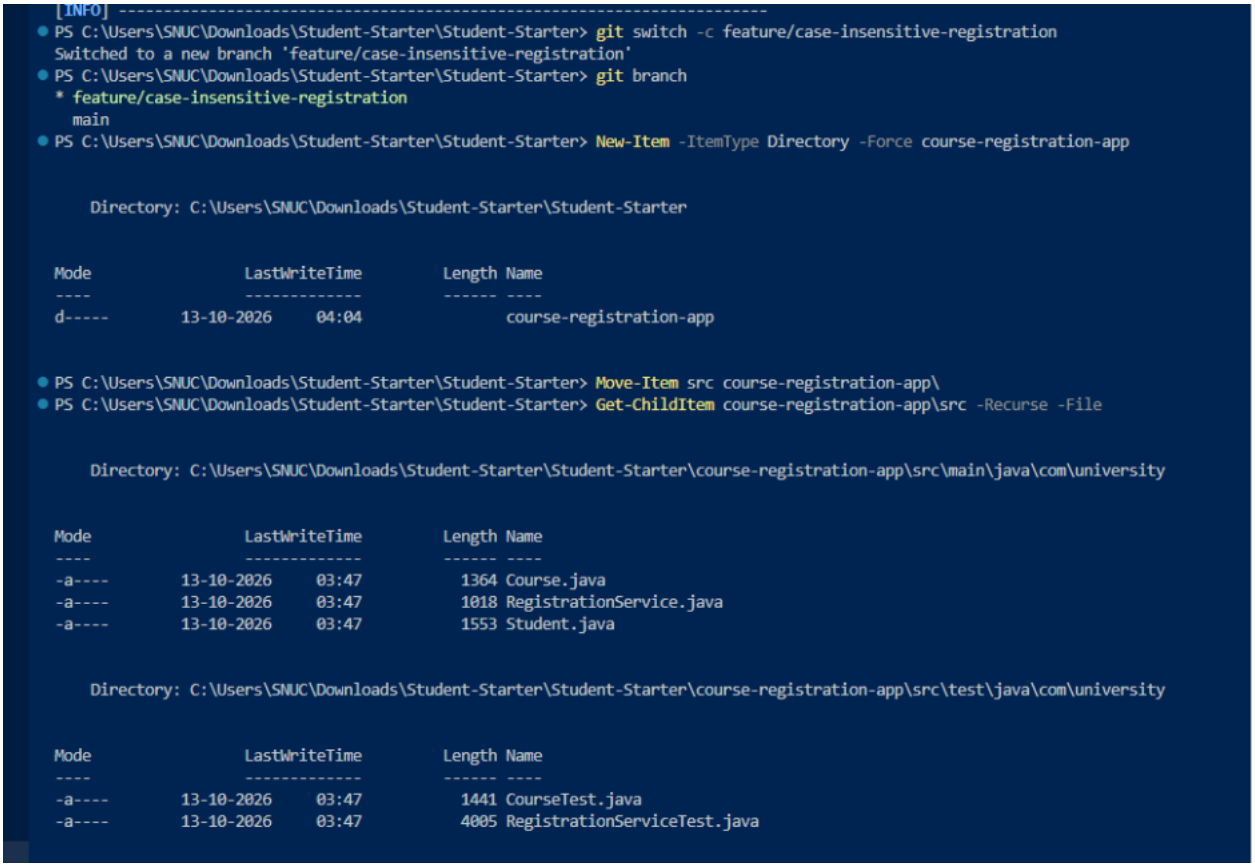
- Do not commit generated `target/` output, credentials, access tokens, or cloud secrets.

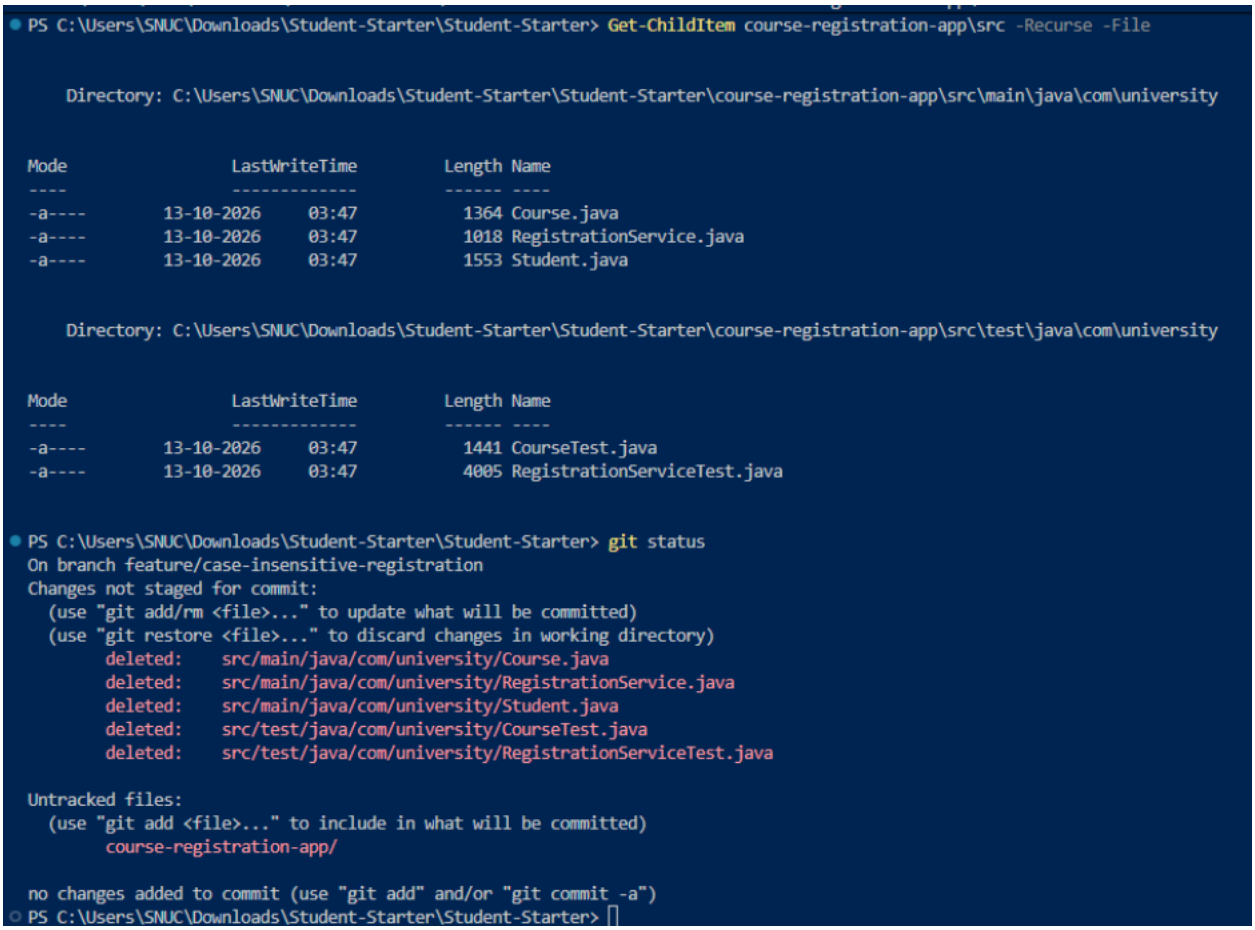
Tasks and marks (common to all sets)

Task 1 — Git workflow (4 marks)

Create a feature branch from the supplied baseline. Make logically separated commits for the defect fix, tests, and your assigned change request. Integrate the completed work into the branch - main

- Show the branch and commit history.
- Ensure the final working tree is clean, apart from any files you explicitly identify as intentionally untracked.





Task 2 — Maven build and test evidence (8 marks)

Objective: Refactor the supplied Course Registration application into a multi-module Maven project using POM inheritance, dependency management and automated testing.

The supplied repository contains a Java application, JUnit tests and an existing `pom.xml`. Refactor the supplied project without recreating the application.

Requirements

a) Create the Parent POM

- Create a parent `pom.xml` at the repository root.
- Set its packaging type to `pom`.
- Define shared properties, including the Java version and JUnit version.
- Declare the application module.

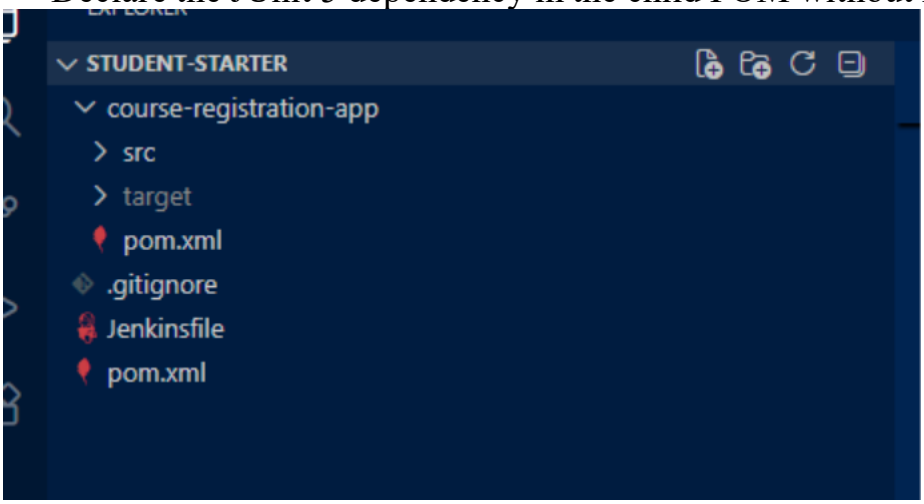
b) Configure the Child POM

- Create a child module named `course-registration-app`.
- Move the existing Java source code and test files into the appropriate directories within the child module.
- Configure the child POM to inherit from the parent POM using the correct parent coordinates and relative path.

- Preserve the existing application functionality and tests.

c) Configure Dependency Management

- Define the JUnit 5 dependency version in the parent's `dependencyManagement` section.
- Declare the JUnit 5 dependency in the child POM without repeating its version.



- Avoid unnecessary or duplicate dependencies.

d) Configure Maven Plugins

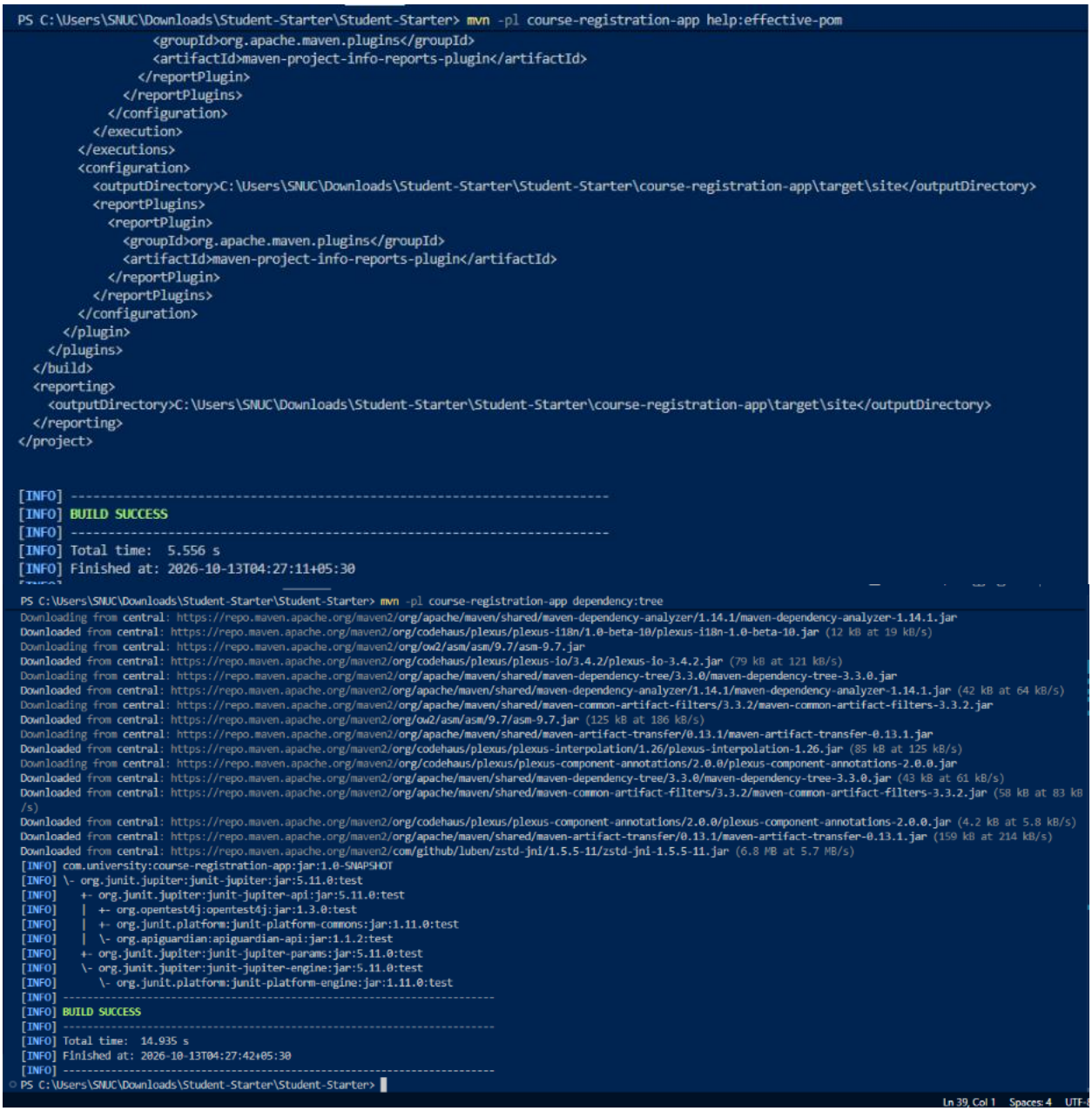
Configure the Maven Compiler and Surefire plugins appropriately for the selected Java version and JUnit 5 test execution.

```
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> git status
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> mvn validate
[INFO] Scanning for projects...
[INFO] ------------------------------------------------------------------------
[INFO] Reactor Build Order:
[INFO]
[INFO] course-registration-parent                                         [pom]
[INFO] course-registration-app                                            [jar]
[INFO]
[INFO] -------------< com.university:course-registration-parent >--------------
[INFO] Building course-registration-parent 1.0-SNAPSHOT                   [1/2]
[INFO]   from pom.xml
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] ---------------< com.university:course-registration-app >---------------
[INFO] Building course-registration-app 1.0-SNAPSHOT                      [2/2]
[INFO]   from course-registration-app\pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] ------------------------------------------------------------------------
[INFO] Reactor Summary for course-registration-parent 1.0-SNAPSHOT:
[INFO]
[INFO] course-registration-parent ......................... SUCCESS [  0.015 s]
[INFO] course-registration-app ............................ SUCCESS [  0.003 s]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.397 s
[INFO] Finished at: 2026-10-13T04:25:55+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter>
```

e) Build and Verify

From the repository root:

- Execute `mvn clean test`.
- Execute `mvn package`.
- Verify that the parent POM successfully builds the child module through Maven's reactor.
- Submit genuine build and test results, including the number of tests, failures and errors.



Expected Project Structure

```
course-registration/
├── pom.xml
└── course-registration-app/
    ├── pom.xml
    └── src/
        ├── main/java/
        └── test/java/
```

Constraints

1. Preserve the supplied application logic and existing test cases.
2. Do not remove or weaken tests to obtain a successful build.
3. The root POM must build the child module.
4. The project must remain compatible with the Jenkins Declarative Pipeline required

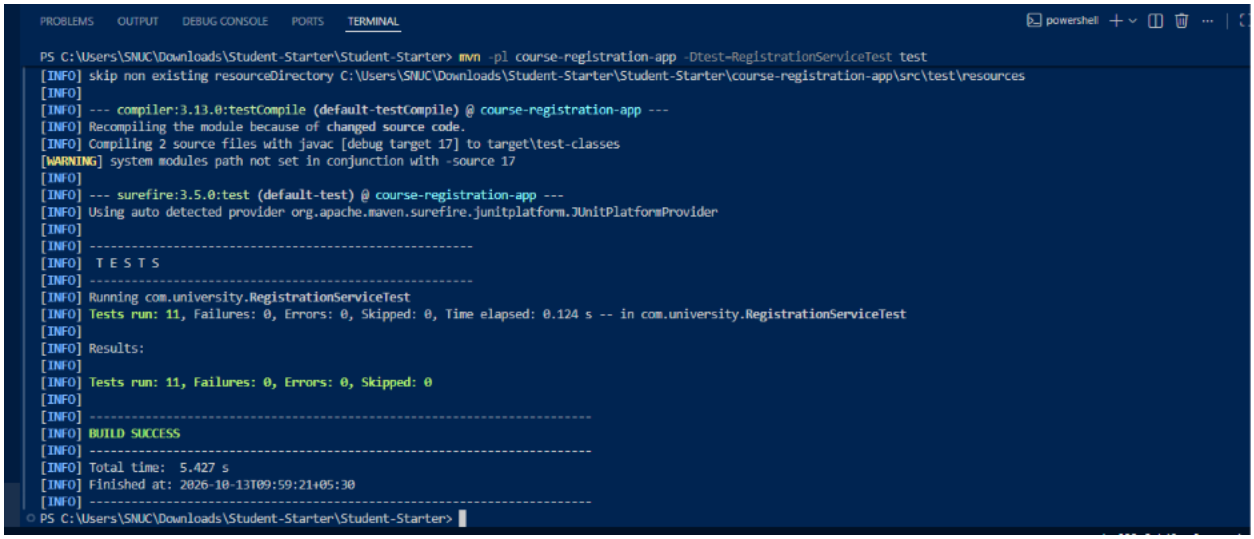
in Task 5.

5. Update the Jenkins artifact and test-report paths to reflect the child module's location.
6. Document the parent-child structure and the build commands in `README.md`.

Task 3 — Diagnose and fix the capacity boundary defect (10 marks)

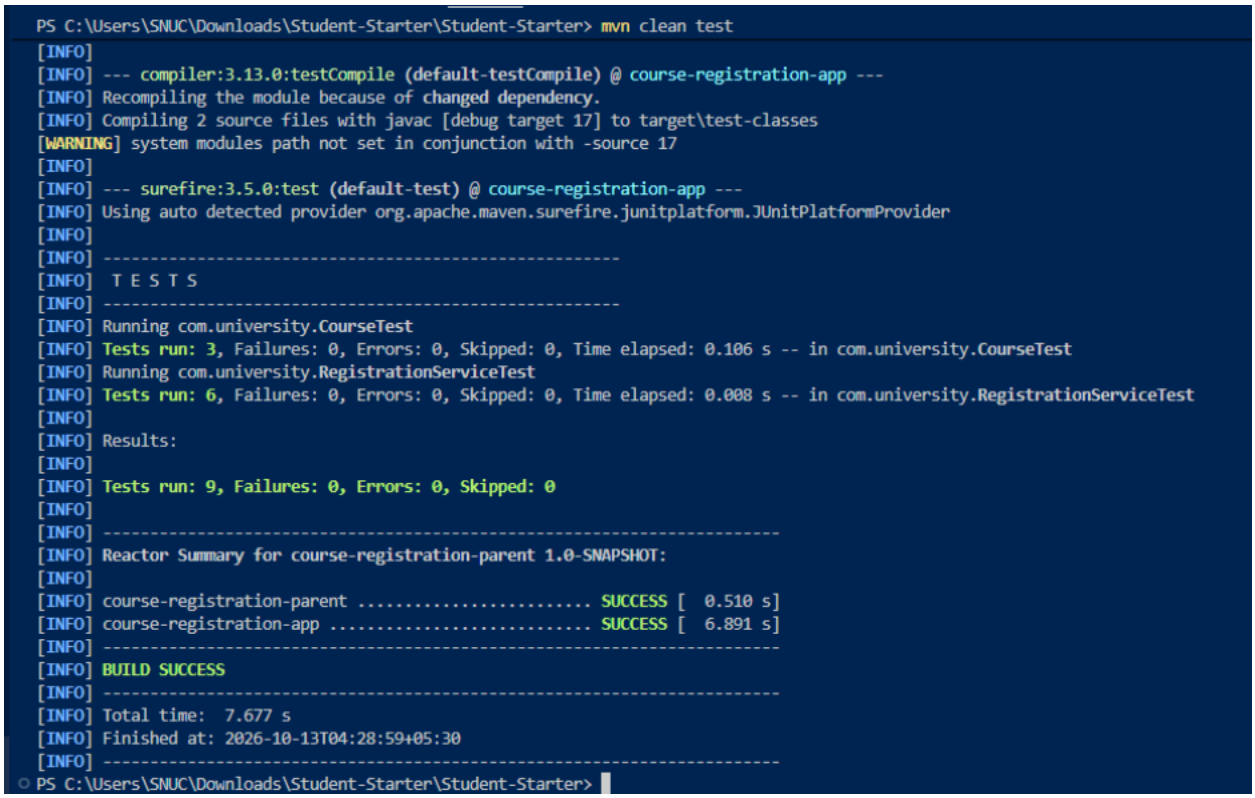
Investigate the registration-capacity logic in the supplied baseline. Add JUnit tests for a course that is already at capacity and for a course with exactly one seat remaining. Correct the defect so that a full course rejects a new registration, while the last available seat can be filled. Verify that a rejected attempt does not change the student's registration/credit state or the course's enrollment count.

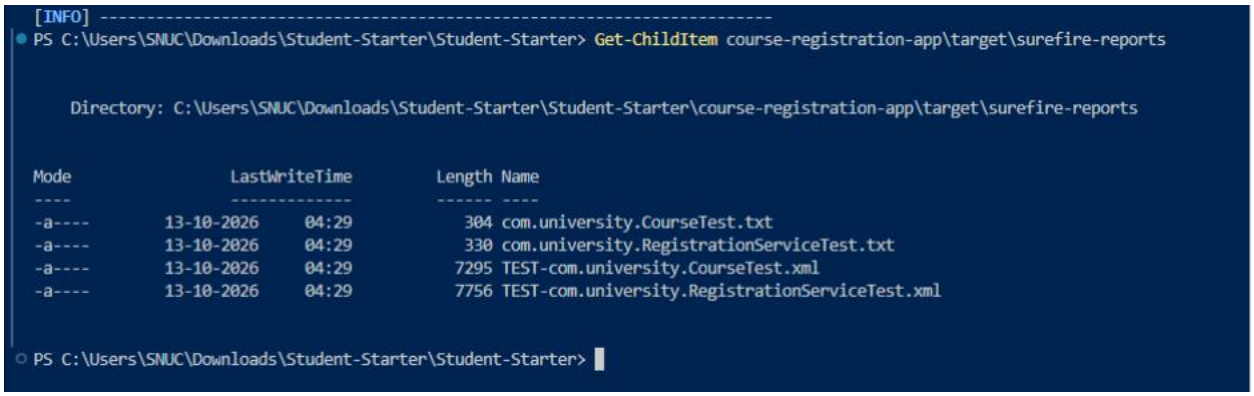
- Include meaningful assertions for both accepted and rejected paths.
- Run the full existing test suite as well as your new tests.
- Do not solve the problem by removing capacity validation.

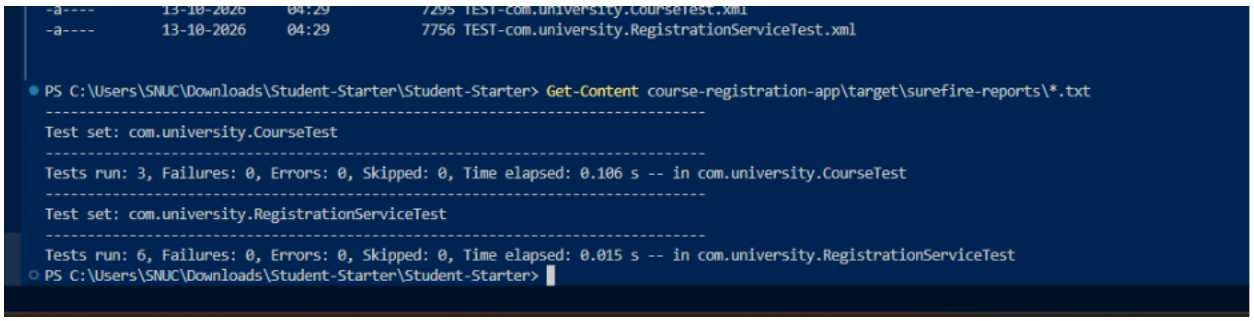


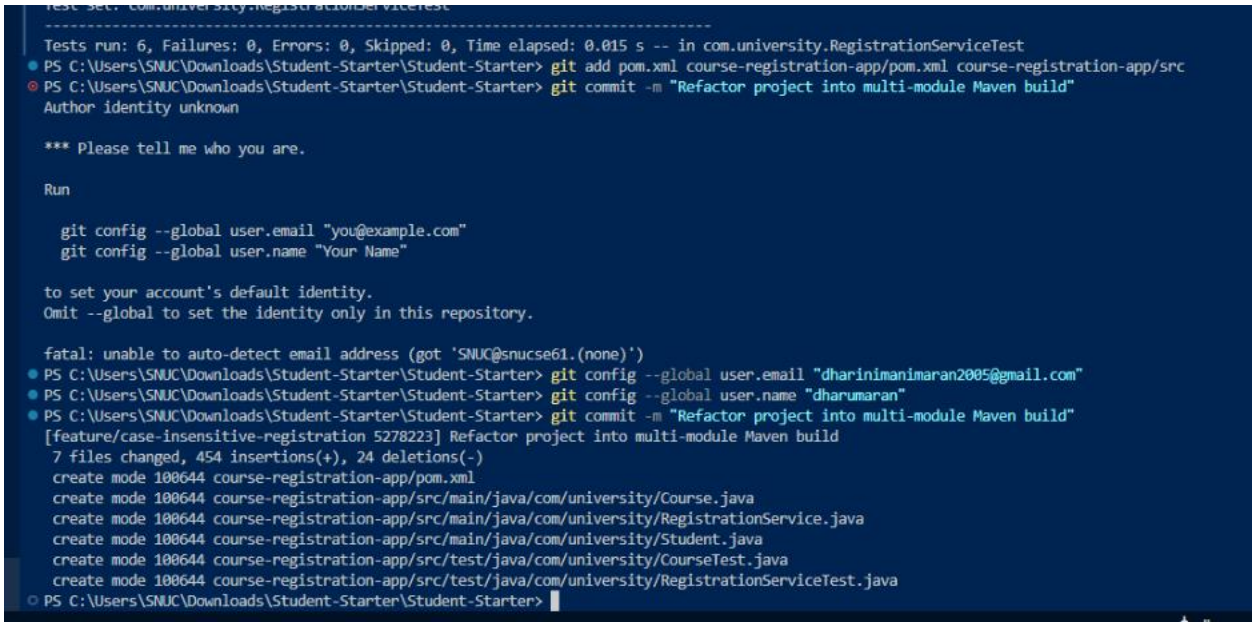
Task 4 — Assigned change request (4 marks)

Complete the specific change request printed on your question set. Implement it and add focused JUnit tests for its acceptance criteria.



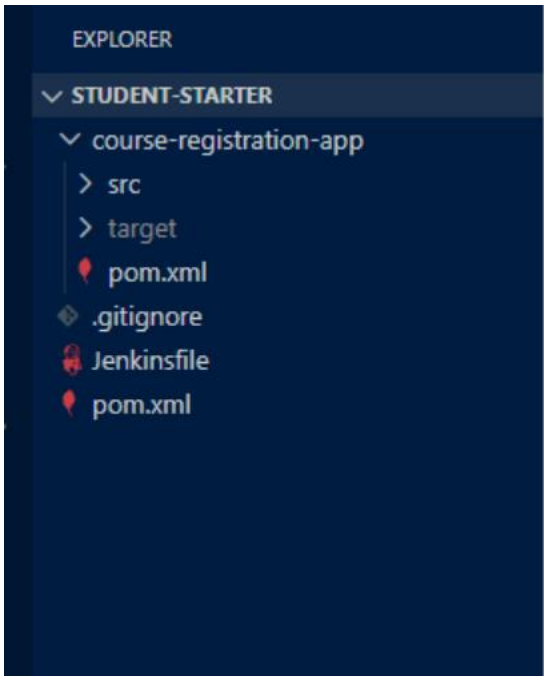






Task 5 — Jenkins Declarative Pipeline (10 marks)

specify that Maven commands execute from the repository root. The expected paths are course-registration-app/target/*.jar for the packaged artifact and course-registration-app/target/surefire-reports/TEST-*.xml for JUnit reports, assuming the child module uses the stated name.



Task 6 — Deploy and Manage the Application using Minikube (8 Marks)

Demonstrate the ability to deploy a containerized application on Kubernetes using Minikube and manage its availability through Kubernetes resources.

```
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> docker --version
Docker version 29.7.2, build a7dcaa6
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> minikube version
minikube version: v1.39.0
commit: 7a9f6a841470a207de8cf4bafcccee0969d8ba10
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl version --client
Client Version: v1.36.1
Kustomize Version: v5.8.1
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> minikube start --driver=docker
😄  minikube v1.39.0 on Microsoft Windows 11 Pro 25H2
✨  Using the docker driver based on user configuration
📌  Using Docker Desktop driver with root privileges
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.51 ...
💾  Downloading Kubernetes v1.37.0 preload ...
    > preloaded-images-k8s-v18-v1...:  347.58 MiB / 347.58 MiB  100.00% 83.78 M
    > gcr.io/k8s-minikube/kicbase:  507.61 MiB / 507.61 MiB  100.00% 13.89 MiB
🔥  Creating docker container (CPUs=2, Memory=4000MB) ...
📦  Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
🔗  Configuring CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl config current-context
minikube
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl cluster-info
```

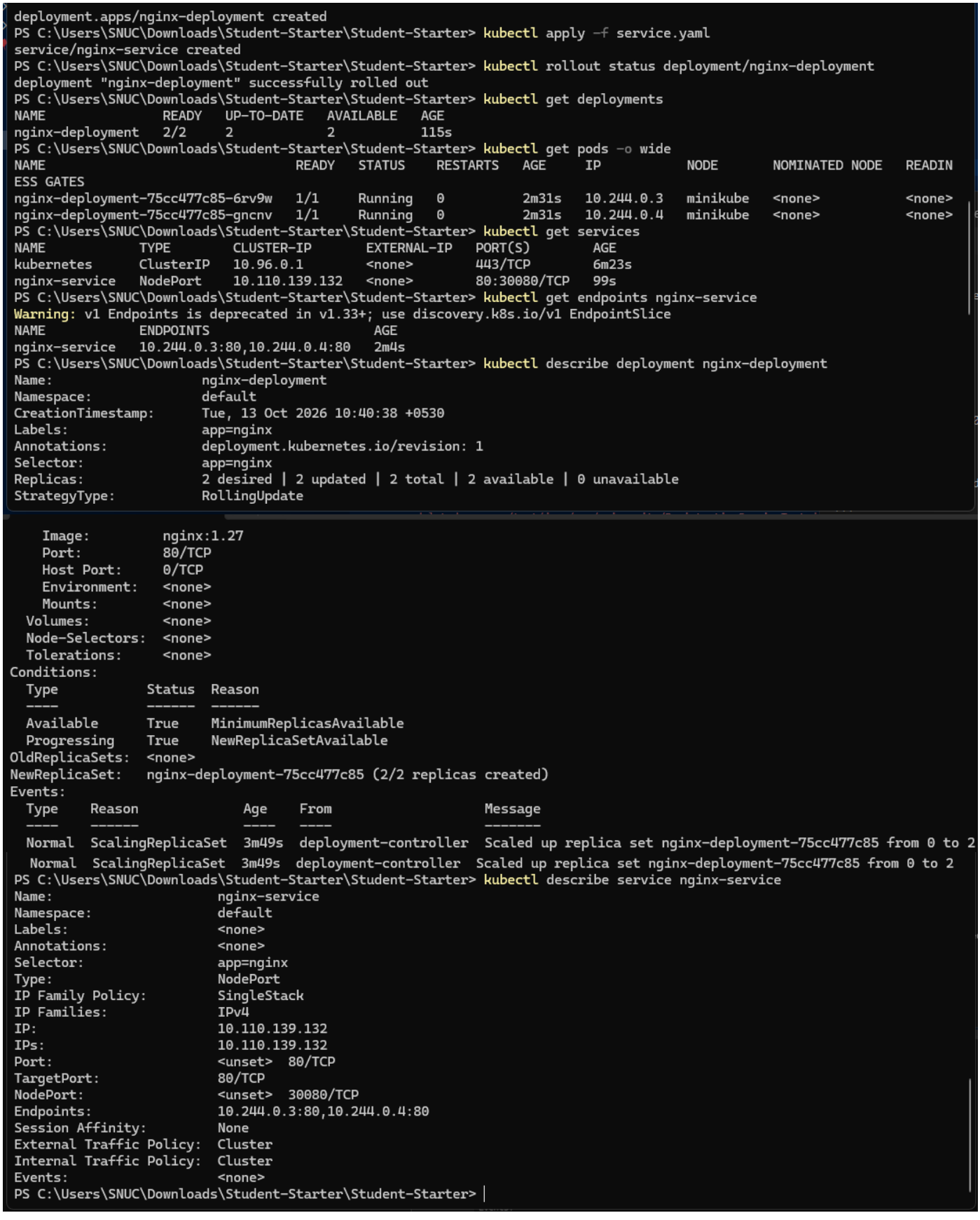
Using your local Minikube cluster, deploy Nginx application and perform the following tasks.

a) Deployment manifest

Create a `deployment.yaml` file to deploy the `nginx:1.27` container image with two replicas. Define appropriate labels and selectors.

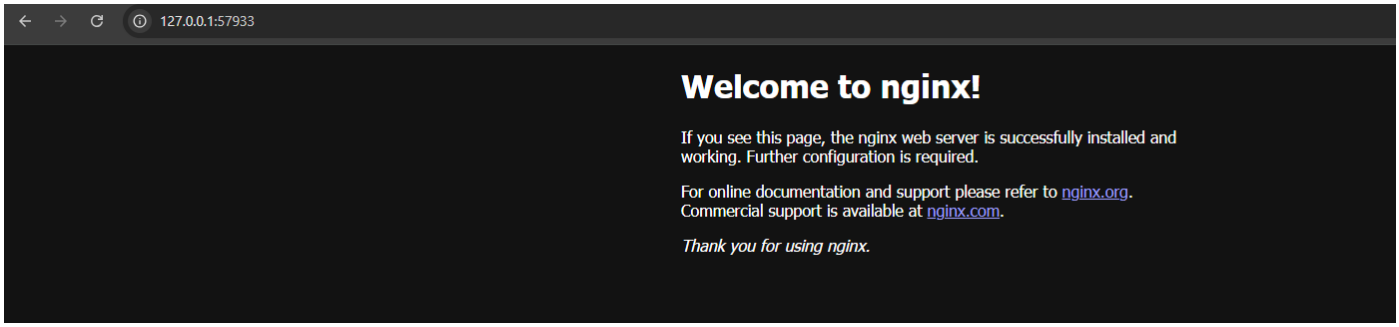
b) Service configuration

Create a `service.yaml` file to expose the Nginx Deployment using a NodePort Service. Verify that the Service selects the intended Pods and that the application is accessible through Minikube.



```
deployment.apps/nginx-deployment created
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl apply -f service.yaml
service/nginx-service created
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl rollout status deployment/nginx-deployment
deployment "nginx-deployment" successfully rolled out
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   2/2     2            2           115s
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE     IP           NODE       NOMINATED NODE   READIN
ESS GATES
nginx-deployment-75cc477c85-6rv9w   1/1     Running   0          2m31s   10.244.0.3   minikube   <none>           <none>
nginx-deployment-75cc477c85-gncnv   1/1     Running   0          2m31s   10.244.0.4   minikube   <none>           <none>
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get services
NAME            TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
kubernetes      ClusterIP   10.96.0.1        <none>        443/TCP        6m23s
nginx-service   NodePort    10.110.139.132   <none>        80:30080/TCP   99s
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get endpoints nginx-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME            ENDPOINTS                     AGE
nginx-service   10.244.0.3:80,10.244.0.4:80   2m4s
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl describe deployment nginx-deployment
Name:                   nginx-deployment
Namespace:              default
CreationTimestamp:      Tue, 13 Oct 2026 10:40:38 +0530
Labels:                 app=nginx
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=nginx
Replicas:               2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType:           RollingUpdate
    Image:         nginx:1.27
    Port:          80/TCP
    Host Port:     0/TCP
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   nginx-deployment-75cc477c85 (2/2 replicas created)
Events:
  Type    Reason             Age    From                   Message
  ----    ------             ----   ----                   -------
  Normal  ScalingReplicaSet  3m49s  deployment-controller  Scaled up replica set nginx-deployment-75cc477c85 from 0 to 2
  Normal  ScalingReplicaSet  3m49s  deployment-controller  Scaled up replica set nginx-deployment-75cc477c85 from 0 to 2
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl describe service nginx-service
Name:                     nginx-service
Namespace:                default
Labels:                   <none>
Annotations:              <none>
Selector:                 app=nginx
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.110.139.132
IPs:                      10.110.139.132
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
NodePort:                 <unset>  30080/TCP
Endpoints:                10.244.0.3:80,10.244.0.4:80
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter>
```

```
Events:                   <none>
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> minikube service nginx-service --url
http://127.0.0.1:57933
 !  Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```



c) Scaling and self-healing

- Scale the Deployment from two replicas to three.
- Delete one running Pod and demonstrate that Kubernetes creates a replacement.
- Verify the resulting Pod count and status.

```
Events:                   <none>
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> minikube service nginx-service --url
http://127.0.0.1:57933
 !  Because you are using a Docker driver on windows, the terminal needs to be open to run it.
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl scale deployment nginx-deployment --replicas=3
deployment.apps/nginx-deployment scaled
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl rollout status deployment/nginx-deployment
deployment "nginx-deployment" successfully rolled out
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get deployment nginx-deployment
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   3/3     3            3           8m41s
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get pods -l app=nginx -o wide
NAME                                READY   STATUS    RESTARTS   AGE     IP           NODE       NOMINATED NODE   READIN
ESS GATES
nginx-deployment-75cc477c85-25h6d   1/1     Running   0          33s     10.244.0.5   minikube   <none>           <none>
nginx-deployment-75cc477c85-6rv9w   1/1     Running   0          8m55s   10.244.0.3   minikube   <none>           <none>
nginx-deployment-75cc477c85-gncnv   1/1     Running   0          8m55s   10.244.0.4   minikube   <none>           <none>
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter>
```

d) Verification and evidence

Submit the YAML files and genuine command outputs demonstrating the Deployment, Service, scaling and self-healing operations.

Use appropriate `kubectl` commands and include a brief explanation of how the Deployment controller maintains the desired number of replicas.

```
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   3/3     3            3           8m41s
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get pods -l app=nginx -o wide
NAME                                READY   STATUS    RESTARTS   AGE     IP           NODE       NOMINATED NODE   READIN
ESS GATES
nginx-deployment-75cc477c85-25h6d   1/1     Running   0          33s     10.244.0.5   minikube   <none>           <none>
nginx-deployment-75cc477c85-6rv9w   1/1     Running   0          8m55s   10.244.0.3   minikube   <none>           <none>
nginx-deployment-75cc477c85-gncnv   1/1     Running   0          8m55s   10.244.0.4   minikube   <none>           <none>
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get pods -l app=nginx
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-75cc477c85-25h6d   1/1     Running   0          92s
nginx-deployment-75cc477c85-6rv9w   1/1     Running   0          9m54s
nginx-deployment-75cc477c85-gncnv   1/1     Running   0          9m54s
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl delete pod nginx-deployment-75cc477c85-25h6d
pod "nginx-deployment-75cc477c85-25h6d" deleted from default namespace
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get pods -l app=nginx -w
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-75cc477c85-6rv9w   1/1     Running   0          10m
nginx-deployment-75cc477c85-gncnv   1/1     Running   0          10m
nginx-deployment-75cc477c85-s7xl7   1/1     Running   0          12s
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter>
```

```
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get deployment nginx-deployment
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   3/3     3            3           11m
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get pods -l app=nginx
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-75cc477c85-6rv9w   1/1     Running   0          11m
nginx-deployment-75cc477c85-gncnv   1/1     Running   0          11m
nginx-deployment-75cc477c85-s7xl7   1/1     Running   0          61s
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get services
NAME            TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
kubernetes      ClusterIP   10.96.0.1        <none>        443/TCP        15m
nginx-service   NodePort    10.110.139.132   <none>        80:30080/TCP   10m
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter> kubectl get endpoints nginx-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME            ENDPOINTS                                      AGE
nginx-service   10.244.0.3:80,10.244.0.4:80,10.244.0.6:80   10m
PS C:\Users\SNUC\Downloads\Student-Starter\Student-Starter>
```

Constraints

- Use Minikube as the Kubernetes environment.
- Use YAML manifests for the Deployment and Service.
- Do not submit fabricated command outputs or screenshots.
- Include the Kubernetes manifests and execution instructions in your submission.

Task 7 — Submission Evidence and Technical Explanation (6 Marks)
Submit a `README.md` containing the following:

1. **Setup and execution instructions:** Provide the commands required to build, test, package and deploy the application.
2. **Git evidence:** Include the feature branch name, relevant commit history and a summary of the changes made.
3. **Maven configuration and build evidence:** Explain the parent-child POM structure, dependency management and Maven build/test results.
4. **Jenkins pipeline explanation:** Explain how the pipeline publishes JUnit test reports and prevents packaging and artifact archiving when tests fail.
5. **Kubernetes deployment evidence:** Include the Minikube Deployment and Service YAML files, relevant `kubectl` outputs, and evidence of scaling and Pod self-healing.
6. **Change summary and limitations:** Summarize the defect correction and assigned change request, list the modified files, and document any known limitations.

Submission requirements: Keep the documentation concise and well-organized. All screenshots, logs and command outputs must be genuine.

QUESTION SET D — Case-insensitive duplicate registration
Assigned change request: Make duplicate-course detection case-insensitive. A student must not be able to register for the same course code twice by changing letter case.

Acceptance criteria

- If a student is already registered for CS101, a second registration attempt using cs101 or Cs101 is rejected.
- A normal registration for a course not already registered remains possible.
- A duplicate attempt does not increase total credits or the course enrollment count.
- Add JUnit tests covering at least two case variations and verifying that state remains unchanged after rejection.

Implementation note

Apply consistent course-code comparison without accidentally making distinct course codes equivalent.

Marks are awarded for correct implementation, focused tests, and evidence that existing registration rules continue to work.